

Jaykumar Goswami

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EDUCATION

Master of Science in Robotics - Northeastern University, Boston, MA December 2026

***Thesis** (Shepherd's Lab): Motorized Back Exoskeleton for Industrial Fatigue Reduction; **Graduate Supervisor** at Makerspace (Trainer for CNC Machine, Waterjet, and 3D Printing); **Teaching Assistant:** Additive Manufacturing*

Bachelor of Technology in Electronics Engineering - BVM Engineering College, Gujarat, India June 2023

***Patent Holder** (Utility & Design); 2x Published [Research Papers](#); First-Class Distinction*

WORK EXPERIENCE

Mechatronics/Robotics Intern - Breaking Inc, Cambridge, MA July 2025 - December 2025

- Designed and built a biotech device integrating optical sensing and embedded control from scratch, iterating through rapid prototyping cycles to deliver lab-ready hardware achieving ± 0.02 OD₆₀₀ accuracy
- Engineered a precision heat chamber for plastic degradation studies - selected materials, designed enclosure geometry, and validated thermal stability at ± 0.5 °C across repeated test cycles
- Integrated sensors, microcontrollers, and closed-loop control algorithms into fully functional lab-ready prototypes, improving experimental reproducibility and cutting data acquisition time by 40%

Robotics and Automation Engineer - BVM Engineering College, India August 2023 - August 2024

- Designed a custom PCB to drive a 14-DOF biped humanoid robot; selected and integrated DC motors with encoders across each joint, designed wire routing and harness layout, and owned full design cycle from schematic capture through board layout and physical bring-up
- Engineered a 360 kg-capacity Automated Storage and Retrieval System with telescopic lifting mechanism owned mechanical architecture from concept through FEA validation, tolerance analysis, and DFM optimization for production handoff
- Led mechatronic development of an industrial AMR (3,300 lb. payload); integrated ROS2 navigation stack with LiDAR-based Cartographer SLAM and high-torque servo drives; deployed system reduced manual material handling by 60% in production environment

PROJECTS

Navigation and SLAM Testing with Autonomous Vehicle | Boston, MA Sep. 2024 - Dec.2024

- Implemented custom Python and C++ ROS2 drivers for an autonomous vehicle, implementing IMU/GPS sensor fusion via Extended Kalman Filter and dead reckoning to enhance localization accuracy
- Validated [Fast-LIO loop closure](#) (GitHub) SLAM on custom-collected Boston Street dataset, generating a high-fidelity 3D point cloud map used to benchmark localization accuracy

IoT-Connected Upper Prosthetic Limb | [Design Patent No.](#) 356336-001 | Gujarat, India Dec. 2020 - Apr. 2022

- Led end-to-end product development of a muscle-controlled prosthetic arm; designed all custom 3D-printed structural components (FDM), engineered a bespoke PCB integrating EMG sensors and microcontroller, iterated form factor for wearability and durability with haptic feedback
- Secured a design patent and awarded \$8,800 for innovative assistive technology recognized for hardware innovation from mechanical architecture through electronics integration

TECHNICAL SKILLS

Mechanical Design:	CAD (SolidWorks -Certified, Fusion 360, Creo), FEA/FEM Analysis, GD&T, DFM & Tolerance Analysis, 3D Printing (SLA, FDM), CNC Machining, Rapid Prototyping
Robotics & Mechatronics:	ROS2, SLAM (LiDAR & Vision Based), Sensor Fusion (GPS, IMU, Stereo Camera, LiDAR), AMR Development, Control Systems, Actuator & Servo Control
Electronics Hardware:	Custom Circuit & PCB Design (KiCad/Eagle), Embedded Systems, Nvidia Jetson Orin, Raspberry Pi, Arduino, Ethernet/IP, OPC-UA, PLC & HMI
Programming & Tools:	Python, C/C++, MATLAB/Simulink, Git, Linux, RTOS, OpenCV